Payday Spire

Concept Document

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Zoud GameLab 2026
"Make FinLit Fun"

Zoud GameLab 2026, "Make FinLit Fun". Web game, English and Arabic, currency AED.

Summary (one page)

One line. Your first year in Dubai as a card battle: twelve months, twelve battles, where bills, surprise costs and sales are the enemies and your money habits are the deck.

What players learn. Pay a card bill in full by its due date, or interest runs on all of it; keep an emergency fund for surprise costs you cannot pay before they happen; card spending lets impulse buys in; pay-later spends future salaries; a bill left unpaid does not go away; pay yourself first (standing order); save ahead for a cost you can see coming (sinking fund); negotiate and compare; profit sharing shares losses too, card interest is charged whatever happens; zakat, the gold nisab and the lunar year, sadaqah and qard hasan.

What is different (3 things).

1. **Credit is the free card in your hand.** Paying a bill properly costs focus, and focus is short. "Put it on the card" costs none. The cost comes back next month as an enemy of its own (the card bill, due in the middle of the month), and every swipe slips an Impulse buy into the deck. Players feel "credit feels free when you are busy" in their hands before they read it anywhere.
2. **You win a life goal, not a boss fight.** Before the year starts the player picks a goal from a young worker's real life in the UAE: money for family back home (AED 6,000), a studio of your own (AED 12,050 to move in: 4 rent cheques, deposit, agent, Ejari, DEWA), or wedding gold (an 18 g 22K set, AED 9,650 with making charge and VAT). The goal has its own jar (a sinking fund), one event of its own and its own cards, and the year ends on "goal reached" or, for example, "AED 970 short", not on a defeated enemy. Finishing a year opens a harder level (lower card limit, bigger surprise costs, a rent rise), so a second year is a new money problem, not the same fight.
3. **A classroom system.** One class code puts the whole class through the same year. Each player copies a result code; the teacher dashboard (`teacher.html`) turns the codes into an anonymous class summary with the three topics to discuss first. No server, no accounts, no names.

Why a deck-builder fits money habits. In a deck-builder the deck is what you have built up, and every card you add keeps coming back. Money habits work the same way. Each rule below is in the game now:

Deck-builder rule	In Payday Spire	What it teaches
The deck is built over time, one card a month	Each month you add one habit card (a standing order, Compare prices, Save ahead)	A habit chosen once keeps paying, or costing, every month after
Cards cost energy, one card costs none	Paying a bill properly costs focus; Put it on the card costs 0	Credit feels free when you are busy
Deck clutter: bad cards get shuffled in	From month 2 each swipe slips an Impulse buy into the draw pile	Card spending makes the next impulse buy easier
Damage comes back next fight	This month's card spending is next month's enemy, the card bill, due mid-month	Pay it in full by the due date, or interest runs on all of it

Deck-builder rule	In Payday Spire	What it teaches
Block that stays	The emergency fund is the only shield that does not vanish at turn end	A surprise cost becomes a plan instead of card debt
Enemies show their next move	Bills, surprise costs and sales show what they will do and when	Plan for costs you can see coming (the flight is announced 3 months ahead)
Removing a card is an upgrade	The coach drops a card for good, for example an Impulse buy	Dropping a bad habit is worth as much as adding a good one
Exhaustion has a cost	Mood at 0: AED 1,500 of tired spending, one card fewer next month, an Impulse buy joins the deck	Rest and planned joy are part of a budget

Judges' 2-minute path (open the files directly, no install or network):

1. Open `index.html?code=K7M3QX`. Press **Start my year**: the battle opens at once, and one glowing card at a time shows what to play (first card in about 7 s by `tools/first_card.py`). Then press 1 (or tap the first card) twice and E to end each turn, then read the month-end summary and answer the quick guess (or skip it). (about 60 s)
2. On the title screen press **For teachers: class dashboard**, then **Load a sample class (20 students)**: class summary, three topics to discuss, guesses before and after, one row per student. (about 30 s)
3. Press **Partner theme preview** (or open `index.html?brand=demo-bank`) to see the white-label frame. (about 15 s)

Plan and budget. 12 months, five paid milestones (Submission, Game Design Document, Playable Alpha, Playtest Beta, Commercial Launch); total USD 66,110 (AED 242,789), itemised in budget.md, with spending never ahead of payments received. A full year plays in about 11 to 13 minutes plus 3 minutes on the end screen (estimated by `tools/playtest.py` from the real buttons and a reading speed of 230 words a minute), so a 45-minute lesson fits two years and a class talk.

Does the lesson hold in the numbers? On 8 class codes (`node tools/regress.js`): steady habits average AED 21,594 with no interest or fees; no emergency fund ends at 50 to 84% of that (interest, cash advance fees and burnouts on every code); every bill on the card 81 to 93%; doing nothing and impulse habits freeze the card on every code. The build fails if no emergency fund ever reaches 90% of steady, or if the end screen says "AED 0" or gives one cause another cause's interest. These numbers are written into this document by `tools/doc_numbers.js` from the same run.

More to try

- **Same code, your last run** (kept in this browser only, up to 20 codes): play `index.html?code=K7M3QX` to the end, then **Play the same year again**. A "Your last run" chip joins the status bar, each month-end summary has a line for it, and the end screen adds a third bar.
- **Goals and levels:** pick Wedding gold under the class code and play a year without a code: the gold jar chip sits in the status bar, the jar card comes into the hand once the month's bills are paid, month 11 brings the gold souk offer and "Ask the making charge". The end

screen tells the goal's verdict, shows achievements and opens level 2. A class code always plays level 1; `index.html?code=K7M3QX&goal=studio` gives a whole class the same goal.
- **Arabic:** `index.html?lang=ar` (full right-to-left layout, Western digits, amounts kept left to right).

Core loop

Start. Type a 6-character class code or open a link with `?code=`; the same code gives the same months, the same surprise costs in the same order, the same fund returns and the same card offers. Month 1 is the move-in month and a practice battle: the first salary pays 24 days (AED 1,700 cash after living costs), and the agent fee (AED 1,100) and the room deposit (AED 1,800, returned when you move out, so it counts in net worth) cost more than that. Three card kinds, taught by doing: a first-time player goes straight to the battle, where one card glows at a time with a one-line guide (Pay from cash on the Agent fee, then Put it on the card, then the card bill it makes shows in place as next month's enemy), and the result shows on the problem and the status bar at once. A second run on the same device skips it. The title screen shows a card bill and three cards of the first hand, so the card battle is visible before Start. Everyone meets Put it on the card in month 1 and the card bill in month 2.

The year's goal and level. Under the class code the player picks the year's goal (the first one is marked, so a first-time player just presses Start): money for family (AED 6,000 sent home in December), a studio of my own (AED 12,050 to move in) or wedding gold (AED 9,650). The goal has a jar in the status bar. Once a month's bills, card bill and surprise costs are dealt with, the jar card comes into the hand (one transfer a turn), and each month-end window offers the jar (marked once the emergency fund holds its goal). Each month intro says what is in the jar and about how much a month gets there by December: the sinking fund sum. Each goal brings one event of its own: a hospital bill at home in month 5 (the emergency fund pays it, or the jar does), a flatmate moving out in month 9 (your share of the room +AED 250 a month unless you find someone: Negotiate), the gold souk offer in month 11 with "Ask the making charge" (7% instead of 12%, AED 430 off). When savings run out, the jar pays next and the end screen says so. Finishing a year opens the next level: 2 card limit AED 8,000 and surprise costs +25%; 3 limit AED 7,000 and the shared room +AED 200 a month from month 4; 4 limit AED 6,000, surprise costs +50%, room +AED 300. A class code plays level 1. `tools/regress_goals.js` plays every goal x level on 30 codes x 11 play styles: steady habits stay first on average everywhere, the card never freezes for them, and the verdict is checked against the jar.

Each month is one battle of three turns.

- **Payday:** AED 7,500 in; rent, food and transport (AED 4,300) go first; AED 3,200 is cash to fight with.
- **Each turn:** draw 5 cards, spend 3 focus. Every card says its result in plain words ("Pay from cash: Bill or surprise cost -600, or the whole card bill, from cash"). With cash in hand, the last turn and the card bill's due turn always hold a Pay from cash, so a bill is never late because of the draw. From month 2 a fresh hand holds at most two copies of one card while other kinds wait in the piles, so a hand is never four Pay from cash and nothing to choose (`tools/regress_design.js` H1).
- **Enemies act at turn end,** on the move they showed as an icon and a sentence: a subscription grows, a surprise cost hits on its turn, a sale slips Impulse buys into the hand.

Some bills arrive at turn 2 and show greyed until then, so the early months keep every player busy.

- **The card bill:** last month's card spending is an enemy, due at the end of turn 2. Paid in full: no interest. Less: interest on the whole bill. Under the minimum (5% of the bill, at least AED 100): an AED 240 late fee too. After a bill not paid in full, the next bill has no interest-free period (interest even if paid in full), as Emirates NBD and FAB terms say; the card forecast and statement line then show that interest instead of promising none.
- **Month end:** bills still unpaid stay owed with an AED 100 late fee and come back next month; this month's card spending closes into next month's card bill; leftover cash goes to the emergency fund, savings or fun (or fills the fund to its goal and saves the rest); the summary lists every line that changed the money.
- **Then:** pick one habit card (or skip). After months 3, 6 and 9 the same window also offers a skills course (paid from savings, then from emergency fund money above its one-month goal) or a coach who drops a card (one pick). Three windows a month in all: the month intro, the month end, the habit.

When	What joins the year
After month 1	Two quick guesses (what the next card bill adds when AED 200 of an AED 2,000 card bill is left; is zakat due on AED 20,000), skippable, asked again at year end with new numbers.
Month 2 on	Each Put it on the card slips an Impulse buy into the deck (people spend more with cards). Ramadan brings planned joy cards (Iftar groceries, Give sadaqah).
Months 4 to 6	**Flight home announced 3 months ahead** (AED 2,200). Save ahead puts money aside for it only; leftover goes back to savings after the flight.
Months 4, 5, 10, 11	Two of these four, moved by the class code: a car AC repair (AED 3,200), a toothache (AED 2,800), a cracked phone screen (AED 900), a rent raise. The two big surprise costs never come back to back.
Month 8	**Payroll delay** on every code: the salary comes two weeks late, after the card bill's due date. The emergency fund pays rent and living costs; without it, rent is a cash advance (fee and interest from day one) and the rest goes on the card. The market dip comes too (hold or sell the fund).
Month 9 / 10	Mega sale with a pay-later phone already in the hand / a friend asks to borrow AED 1,500.
Month 12	The year-end sale rush (the final battle), then the quick check and the end screen.

Rules worth stating.

- **Emergency fund:** the deck-builder "block" that does not vanish at turn end. A surprise cost shows the turn it is due, and the player chooses how to pay it before then: Emergency fund pays (a 0-focus card in the hand while the fund has money; on the card bill it pays only what cash cannot cover), Pay from cash (the month's bill money), or Put it on the card (free if the next statement is paid in full, but each swipe lets an Impulse buy in). Whatever is unpaid when it is due goes on the card, with a worry when the fund could not have covered it (Mood -1 for each AED 300 over what the fund holds, at most -8): the fund sits in its own account and pays only when you move the money. It also pays rent and living costs when the salary comes late. With the fund empty, savings can pay (Break into savings, and automatically for the late rent), but each time a savings goal is set back (Mood -3) and the end screen names it.

- **Mood:** falls with work, side gigs (one a month, Mood -8) and debt worry; a full emergency fund (one month of costs) softens the monthly fall, but a card balance carried past its due date counts against it. Under 12 at a month's start: AED 400 of tired spending and an extra Impulse buy that month. At 0 the player burns out: AED 1,500 of tired spending at once, one card fewer next month, an Impulse buy joins the deck, and the year cannot be graded Steady year. Treats and planned joy keep it up, so skipping them to work more does not pay. An Impulse buy can be walked away from (1 focus, Mood -3).
- **Card limit:** over AED 10,000 (lower at levels 2 to 4) the card is frozen and the year ends.

End of the year. The quick check returns with new numbers (an AED 3,000 bill paid in full; AED 50,000 held a lunar year). Then the end screen: grade (Steady year, You made it, Card frozen; under You made it, the reasons this year had: card debt, an overdue bill, a card bill not paid in full, a thin emergency fund or a burnout), net worth, interest and fees, emergency fund in months; **three sentences on why the year went this way**, written from what happened; **Before and after** ("Right answers: x of 2 in month 1 → y of 2 at year end", with the worked answers tied to this year); **the same year with steady habits** as a bar beside yours, with one line on whether each of you reached the goal (or how much short); a zakat check against the gold nisab; the **goal's verdict** (reached, or what the jar holds and how much is short, what the jar paid when savings ran out, and, on a missed goal that savings and the emergency fund could have covered, what moving them would have done: spare money reaches the goal with the fund whole, breaking the fund reaches it but leaves less than a month of costs for the next surprise), **achievements** (no interest or fees, emergency fund full, no Impulse buy slipped through, goal reached, level finished; new ones marked) and the next level; and a **result card** (net worth, interest and fees, emergency fund, Impulse buys, card bills not paid in full, bills left unpaid) copied as text with a **result code** (`PS2-...`: the year's numbers, the goal, the level and whether the goal jar was full). **Discussion cards** print two questions for each problem and decision this player met.

Teacher dashboard (`teacher.html`). Paste the copied results; only the result code line is read, and two check letters catch a wrong or swapped letter. It shows class numbers (steady years, frozen cards, card debt, net worth, emergency fund, interest and fees, Impulse buys, sadaqah, the share who reached their year goal), grades, the year goals and levels (for each goal: students, reached, the share reached and the average in the jar), the month 1 guesses and the before-and-after gain, the **three topics this class found hardest** (core money topics first; single decisions only fill a free slot) with the discussion questions, and one row per result. It prints, and nothing is sent or saved. The sample class is 20 made-up students who played the real year of code SAMPLE on the game rules with careful, mixed or impulsive habits, each with one of the three goals at level 1 (`tools/sample_class.js`): the sample class reached 25% of its year goals (5 of 20; money for family 3 of 9, a studio 0 of 4, wedding gold 2 of 7). A teacher can give the whole class one goal with the game link (`index.html?code=CLASS7&goal=studio`); a class code always plays level 1. Codes from an older copy of the game (`PS1`, no goal or level) are still read and shown apart. Percentages stay "50%" in Arabic too: the sign sits inside the number's left-to-right isolate (`PS.pct`), checked on every Arabic screen of the game and the whole dashboard.

Cards and enemies

Group	Examples	What it teaches in play
Starter cards	Pay from cash, Put it on the card, Top up emergency fund, Wait 24 hours, Small treat	Pay now or later, defend, cool off, rest
Habit cards (pick one a month)	Standing order to savings, Bills on autopay, Compare prices, Negotiate, Say no to pay-later, Pay down the card, Costliest debt first, Budget check, Save ahead, Pack lunch, Take the metro, Weekend side gig, Profit-sharing fund, Give sadaqah, Cash advance	Habits are what you build a year with; some help, some cost later
Planned joy	Iftar groceries, Eid gifts	Religious months are good months, planned for
Burdens	Impulse buy (costs AED 200 if still in hand), Pay-later reminder (takes a hand slot)	What card spending and pay-later leave behind
Enemies	Bills (agent fee, room deposit, moving van, phone, subscriptions that grow, a traffic fine, car registration, a Salik toll top-up, DEWA in summer, trial subscriptions, a car service, extra baggage in the flight month), surprise costs (car AC, toothache, cracked screen, glasses, laptop, flat tyre), a rent raise, sales (flash sale, new phone, mega sale with pay-later, November deals, year-end rush), the card bill	Money problems, each with its own rule. Each kind comes with one answer card in the first hand of its month (Compare prices, Wait 24 hours, Negotiate) when the deck has none, so every month has a decision

Target audience

Main: young adults aged 18 to 25 in the UAE and the Gulf on their first salary (new graduates, first-job workers, new expat arrivals; AED 7,500 a month fits a first graduate job). Secondary: classes aged 15+ and university or employer onboarding. English with an Arabic toggle, short sentences, every card a result sentence.

Learning goals per mechanic

Mechanic	What the player learns by doing it
Put it on the card costs 0 focus, the card bill comes back as an enemy	Credit feels free when you are busy; the bill comes back when next month's problems arrive.
Card bill paid in full: 0; less: interest on all of it; under the minimum: a late fee	Pay the card in full by the due date. Paying most of it is not enough.
Each swipe adds an Impulse buy to the deck	Paying by card makes it easier to spend more.
Surprise costs are due on a set turn; the player picks fund, cash or card	Using the emergency fund for an emergency is a decision, not magic: a full fund turns a shock into a non-event, cash leaves bills short, the card is free only if paid in full and lets impulse buys in.
Bills left unpaid come back with a late fee, month after month	A bill you skip does not go away; it grows.

Mechanic	What the player learns by doing it
Payroll delay in month 8: the emergency fund pays rent and living costs; without it, rent is a cash advance	An emergency fund is for late pay too, not only surprise costs; a cash advance costs a fee and interest from day one.
A bill not paid in full takes away the next bill's interest-free period	One short payment costs more than one month of interest.
Pay-later phone: AED 900 today, AED 900 from each of the next three salaries, a reminder card in the hand	Pay-later spends future salaries; each month starts smaller.
Standing order card	Pay yourself first; automate saving.
Save ahead for the flight home, announced 3 months ahead	Save ahead for a cost you can see coming (sinking fund).
Wait 24 hours, Say no to pay-later	An impulse cools off; saying no costs a little mood, not money.
Negotiate, Compare prices	Recurring costs can be talked down; prices differ.
Leftover cash: emergency fund, savings or fun; Mood meter	Fun is part of a plan; too little leads to burnout; debt costs worry too.
Profit-sharing fund with a dip, hold or sell (2% exit fee)	Profit and loss are shared (no riba). Selling in a dip makes the loss final.
Friend's loan from savings, or as a cash advance	Qard hasan returns the same amount; a cash advance costs a fee and interest from day one.
Zakat check, nisab bar, sadaqah lines	Zakat al-mal is 2.5%, due only on wealth above the nisab held for a lunar year (gold nisab, 85 g; some scholars use the lower silver nisab). Below it, giving is sadaqah.
Quick guesses, end screen, Your last run, class code, dashboard	Predict, see evidence from your own year, check again; change one habit and compare; talk about money as a class.

Why these features

What is different. We read the public descriptions of the 53 jam entries (10 Nov 2026). Most budgeting entries split a salary into jars or envelopes; none we read is a deck-building battle. Here a deck-builder's own rules carry the money ideas: block that persists is the emergency fund, a card that costs no energy is credit, a card that clogs your hand is pay-later, a curse in your deck is the Impulse buy that card spending let in, and the enemy's announced move is a bill's due date.

Why it should help learning.

- **There is room to improve.** In PISA 2022, 15-year-olds in the UAE scored 441 in financial literacy against an OECD average of 498 (OECD, PISA 2022 Results Volume IV).
- **Financial education works, modestly.** A meta-analysis of 76 randomized experiments found effects of about 0.20 standard deviations on financial knowledge and about 0.07 to 0.10 on behaviour (Kaiser, Lusardi, Menkhoff and Urban 2022, NBER WP 27057). We do not claim more than that for a game.
- **Just in time.** Classroom effects fade; teaching right before the decision helps (Fernandes, Lynch and Netemeyer 2014, Management Science). Each rule appears on the enemy at the

moment it matters: the card bill states its interest before the due turn, a surprise cost states its turn and who pays.

- **Card payment raises spending.** Willingness to pay was higher when buyers paid by credit card (Prelec and Simester 2001, Marketing Letters), and card payment raised impulsive purchases (Thomas, Desai and Seenivasan 2011, Journal of Consumer Research). The game turns this into one Impulse buy per swipe.
- **A cash cushion matters for wellbeing.** Money in the bank predicted life satisfaction more than income or debt (Ruberton, Gladstone and Lyubomirsky 2016, Emotion); scarcity and stress push short-sighted choices (Haushofer and Fehr 2014, Science). Hence the worry from a surprise cost that goes on the card, and burnout spending.
- **Commitment saving is the standing order.** A savings account the saver could not easily touch raised savings (Ashraf, Karlan and Yin 2006, QJE). The standing order card moves money to savings on payday, before the battle starts.
- **Active learning.** In a field experiment, active learning raised saving where a lecture did not (Kaiser and Menkhoff 2018, DIW DP 1743). Every lesson here is a card played, not a paragraph.

Do the numbers reward the lesson? A teaching game fails if the habit it preaches is not the one that wins. `node tools/regress.js` plays five habit styles through the same engine as the browser on 8 class codes. Average net worth after the year: steady habits AED 21,594 with no interest or fees; every bill on the card, paid in full when possible, AED 18,883 (87%, more Impulse buys, and a card bill that lands in the payroll delay); no emergency fund AED 16,076 (74%, 50 to 84% by code: the payroll delay becomes a cash advance, surprise costs go on the card and the next bills are not paid in full, burnouts follow); doing nothing below zero, card frozen on every code; impulse habits below zero, card frozen in months 8 to 10 on every code. The test fails the build if steady stops ending highest on any code, if no emergency fund ends above 90% of steady on any code, or if the end screen of that year does not name the cash advance or the surprise cost among its three reasons. It also plays every month of 12 codes and fails if a month has no decision (two moves on the problems, or a move that competes with earning or saving for focus). Each reason sentence is built from clauses and drops a clause worth AED 0; interest is traced to the card money that caused it (a ledger shares each charge in proportion), so no cause is blamed for another's interest.

How it maps to the rubric

Criterion (weight)	Where to look
Gameplay (25)	A short turn loop with announced enemy moves, rising pressure (summer bills, the mega sale, the year-end rush), a deck you build and prune. Full 12-month runs with no console errors (`tools/playtest.py`, `tools/shots_v3.py`).
Financial literacy (25)	Every card and enemy is a money rule; interest, late fees, limits, risk and giving come from the rules every month. See the learning goals table and the appendix.
Educational impact (15)	Consequences on screen when they happen, Before and after, the same year with steady habits, discussion cards, and the dashboard's three hardest topics.
Creativity (15)	Deck-builder rules mapped to money (Summary and Why these features); Your last run.

Criterion (weight)	Where to look
Development and commercial (20)	Both paid products already run (`teacher.html`, `?brand=demo-bank`); a school pilot kit is ready (one-page pilot plan, dashboard CSV and pilot results page); revenue in four stages with entry conditions; cards, enemies and months are data tables; itemised budget and five milestones (budget.md).
Localization (+5)	AED, Arabic UI, Ramadan and Eid as planned joy (never enemies), sadaqah, qard hasan, profit sharing set against interest, a gold nisab zakat check, the agent fee, DEWA in summer, the metro, a flight home for expat workers, a maqam Hijaz music loop.
Accessibility (+5)	See below.

Commercial plan

- **Free web version:** the reach channel, with class codes, discussion cards and the dashboard.
- **School and university licence:** teacher mode and more scenario packs, built on `teacher.html`. About AED 1,500 per school per year; free pilots first (the 4-week pilot below).

How the free lesson leads to a school licence. What a school would pay for is the lesson around the game, not the game itself. Today the free version already holds one complete 45-minute lesson: `docs/lesson_plan.md` (the same steps are in `teacher.html` under Plan the lesson), a class code that gives every student the same year, a year preview the teacher reads the day before, a class summary made from pasted result codes, and five discussion questions with the points of a good answer. A school that runs it once has a lesson it can repeat with any code. The licence adds what a school needs for a term rather than one lesson: more scenario packs (student year, young family), the teacher mode of phase P3 with one code per class and saved class reports with no personal data, and a pre/post check for the school's own records. The free lesson is how a school tries it; no school has bought a licence yet, and the price above is an assumption.

- **White label for banks, Islamic banks and fintechs:** their name, logo, colours and later their own product cards (a savings account card, their card's real terms) for youth accounts and CSR programmes (`?brand=demo-bank`, fictional). Priced per project (setup plus yearly maintenance).
- **Employer onboarding:** new-hire and expat-arrival packs ("your first UAE salary"), per company.
- **Scenario packs:** cards, enemies and months are data, so a pack is a new deck and enemy set: student year (allowance, part-time job), freelancer year (irregular pay), young family (school fees), debt payoff (smallest first against highest interest first).

Market evidence (what others already do; no partner has signed with us):

- **A school channel exists.** Zoud and Alef Education reported a digital financial literacy programme for grades 5 to 12 in 111 schools with 17,841 students, with 72 lessons in English and Arabic (TechAfrica News, 10 Jun 2026). Payday Spire fits as an activity module next to the lessons.

- **UAE banks already pay for school financial education.** Dubai Islamic Bank reported campaigns in about 50 schools reaching up to 4,000 students (DIB, 7 Nov 2023); Emirates NBD ran a web course in 10 schools for about 300 students (Gulf News, 14 Jun 2021); Emirates Islamic worked with 4 government schools and 120 students (Emirates Islamic, 15 Oct 2025).
- **Price precedents are from the US market, not the UAE.** PersonalFinanceLab lists sponsored school licences from $895 a year (one class) to $5,995 a year (1,000 students), and a white-label budgeting or stock game at $12,000 to $20,000 a year (PersonalFinanceLab); Stock-Trak lists sponsor packages from $1,425 (100 accounts) to $12,000 (1,000 accounts) (Stock-Trak). Our AED 1,500 per school is an assumption checked against these, not a quote from a UAE buyer.

Who we plan to approach first (with Zoud introductions where possible): private schools in Dubai (KHDA) and Abu Dhabi (ADEK) with financial-literacy or life-skills classes; university career centres; Islamic bank youth and CSR teams; card issuers and fintechs serving first-job customers, including pay-later providers; employers that hire many graduates or new expat staff. Targets are in the Roadmap.

Local work in the UAE. The budget pays for playtests in UAE classes, a native Arabic reviewer, a Shariah reviewer, and one pilot visit with a local coordinator (budget.md, lines 4, 5, 9 and 15).

Revenue in four stages. Each stage starts only when the one before it has shown something a buyer can check. The conditions and targets are plans, not results.

Stage	Who pays	What they get	Starts when	Measured by
0. Free web lesson (now)	Nobody	The game, the dashboard, the 45-minute lesson, English and Arabic	Live at submission ([ITCH_URL])	Pilot classes that send a results CSV; web plays once privacy-safe counting is added (budget line 10)
1. School term licence	A school, or a sponsor for it	Scenario packs, one code per class, saved class reports with no personal data, a pre/post check	3 pilot classes have run the 4-week pilot and sent their CSVs; at least one teacher says they would run it again (P4)	Paying schools; licensed teachers who run a second lesson in the term; the pre/post gain (target +20 points, Roadmap P4)
2. Bank and fintech white label	A bank's or fintech's youth or CSR team	Its name and colours, later its own product cards; licences for the schools it supports	Pilot results pages from 3 classes to show; Shariah sign-off (P3); privacy policy for school use (budget line 13)	Proposals sent (target 1 in the first 60 days after launch); signed projects; classes reached per sponsor
3. Employer onboarding	Employers that hire many graduates or new expat staff	A "first UAE salary" year for induction	One white-label build shipped and the first-job pack done (P3)	Companies; new hires who finish the year; their card-bill-paid-in-full rate

Pilot kit, ready now. `docs/pilot_onepager.md` (also `dist/Payday_Spire_School_Pilot.pdf`) is the one page for a school or a bank CSR team: the problem, what already works, a 4-week pilot (two lessons on two class codes and one replay), what is measured and how, and the price assumptions with their US sources. The dashboard measures the pilot itself: **Download results (CSV)** gives one row per student with no names (grade, months finished, card bills not paid in

full, emergency fund goal reached, interest and fees, month 1 and year-end answers), and **Open the pilot results page** prints a one-page results sheet in English or Arabic, with the two class codes side by side. No pilot has run yet.

Running it as a team of one: risks and answers.

Risk	Answer	Budget
The developer is delayed or ill	Plain JavaScript with no build step; rules, cards, enemies and months are data; regression checks run in one command; the Game Design Document (P2) lets a contract developer pick up the work	Contingency, line 16
Too much art for one person	A contract 2D artist makes card faces and enemy art	Line 2
Arabic or Islamic finance wording is wrong	A paid native review and a written Shariah sign-off before the Playable Alpha	Lines 4 and 5
The lesson does not teach what it claims	A financial-literacy educator writes the learning goals and the pre/post check; pilot CSVs show the result	Line 6
No one on the ground in the UAE	A UAE local coordinator and one pilot visit; Zoud introductions where possible; the pilot kit and results page as the demo	Lines 12 and 15
Schools buy slowly (once a year)	Free pilots first, sponsors pay for schools, small targets (1 paying school in 60 days)	-
School devices and privacy rules	QA on low-end Android, iOS Safari and school laptops; no names, no server; a privacy policy for school use	Lines 8 and 13

Public links (filled in on submission day): play [ITCH_URL], source [GITHUB_URL], 2-minute video [VIDEO_URL].

Roadmap

Five phases, each ending at a milestone and payment in budget.md (P1 to P5). Targets are goals, not results.

Phase and payment	Months	Deliverable (as in budget.md)	Main work	Targets
P1. Submission	Nov 2026	Game, budget and concept document received	The jam build above (31 card kinds, 31 enemy kinds, 12 months, EN/AR, dashboard, brand demo).	-
P2. Game Design Document	1-2	Game Design Document approved	Learning goals and pre/post check with the educator. Shariah review starts. Paid Arabic review by a real native speaker. Local price check. First partner contacts.	5 schools or partners contacted.

Phase and payment	Months	Deliverable (as in budget.md)	Main work	Targets
P3. Playable Alpha	3-6	Playable Alpha: 3 scenario packs (deck and enemy sets), teacher mode, Arabic and Shariah review applied, new card and enemy art	Packs: student year, first job, young family. Teacher mode on the dashboard (code per class, lesson plan, class reports, no personal data). Card faces and enemy art made for this game. Easy and Hard decks.	2 pilot schools or university groups agreed.
P4. Playtest Beta	7-9	Playtest Beta: 35+ independent testers in 2 rounds, findings report, accessibility audit passed	Two rounds, many testers in UAE classes. Pre/post check in pilot groups. Balance runs for every pack. Accessibility audit. One pilot visit with a local coordinator.	70% of started runs reach month 12; correct answers on the pre/post check rise by 20 percentage points; 3 pilot schools or university groups.
P5. Commercial Launch	10-12	Commercial Launch: stable V1 on at least one storefront, 60-day launch plan	PWA and a store build. School licence and white-label offer. Partner demo kit.	In the first 60 days: 5,000 web plays, 1 paying school or institution, 1 white-label proposal sent to a bank or fintech.

After month 12: country packs (KSA, Egypt, Malaysia) and more white-label builds, paid for by licences. A Payday mode (sort the salary into bills, emergency fund, savings and spending before each battle) is a candidate extension, already tested as the bill-sorting sister prototype Jar Rush.

Team

- **Wonho Lee** (Seoul, South Korea), solo developer: design, code and art direction. Full-stack developer, doing freelance and client work since 2022 (more than 20 projects). Works at NICE D&B on projects for the Korea Creative Content Agency (KOCCA), leading company data and database design.
- **Shipped and used by real users (at NICE D&B):**
- **K-EXPO event web app.** Built on his own and used to run three K-EXPO export business meetings (Mexico City, Chile and Vietnam), each with more than 600 seller and buyer accounts. It automates QR passes, visitor routing and check-in, and replaced an outside vendor.
- **2026 University Comics & Webtoon Contest platform** (tooncareer.com). Built the entry platform and now runs it; about 300 entries were submitted through it.
- Also built an AI rental-law help app for Android on his own.
- **Recent public work you can check:** CartPilot, an AI shopping agent for the PayPal AI Hackathon (Devpost, demo video); Story Behind the Photo for the UGREEN AI NAS Hackathon (Devpost).
- **Time on Payday Spire:** about 80 hours a month for all 12 months, the same assumption as the developer line in budget.md. Progress is shared in a monthly progress report and checked through the milestone deliverables.

- **Backup:** plain JavaScript with no build step; the money rules, cards, enemies and months are data files. If the developer is delayed, a contract developer can pick up the work from the repository and the Game Design Document.
- Planned contractors (see budget.md): 2D artist, sound, Arabic localization, Islamic finance reviewer, financial-literacy educator, QA, and a UAE local coordinator.

Accessibility

Checked on 11 Nov 2026 against WCAG 2.2 AA with `tools/a11y_audit.py` and the scripted playtests: screen reader messages through two live regions (polite and assertive) for turns, card results, month-end summaries and the end screen; dialogs named by their heading and described by their first line, focus kept inside and returned; a visible focus ring on every control; 1,203 text elements on 11 screens in English and Arabic (title, settings, month panel, battle, status bar, statement, quick guess, end screen, discussion cards, partner theme, teacher dashboard) with 0 contrast failures (text over pictures checked by eye); `lang` and `dir` follow the language, with codes and amounts kept left to right. Play options: Okabe-Ito palette with an icon, a border style and a word on every colour cue, captions for every sound in their own line (never over the game), 3 text sizes, reduce motion (follows the system setting), sound and music on or off, full keyboard play (1 to 9 play a card, T or arrows change the target, E ends the turn, H help, Esc closes a window). The battle screen keeps the hand and End turn on the screen at 1280x720, 1280x800, 390x844, 390x560 and 360x640, at all three text sizes, in English and Arabic (checked by `tools/layout_check.py`: no sideways scroll, nothing covered). Still to do: NVDA or VoiceOver on a real device, and the paid manual audit in phase P4.

AI tool disclosure

Built with Claude (Anthropic) as a coding and writing assistant: game code, line icons drawn in code (SVG), synthesized sounds and the music loop (WebAudio), English text and the first Arabic draft. The 27 pictures (title and battle backgrounds, two ending scenes, 19 enemy portraits and 4 card frames) are Payday Spire's own art, generated with Google Nano Banana from the developer's own prompts during this jam (November 2026) and checked and placed by the developer. Phase P3 adds painted card faces and the art for new scenario packs (budget line 2). The developer directed the design, chose the concept, and reviews and tests the build. There are no third-party stock image, audio or font assets. The Arabic text was checked by an AI review written from a Gulf native speaker's point of view; a review by a real native speaker is planned in phase P2 (months 1 to 2).

Appendix: accuracy notes and sources

- **Card interest** of 3% a month is within typical UAE card rates (2 to 4% a month), called interest (riba) to contrast with profit sharing. A card bill is due about 25 days after the statement date (Emirates NBD and FAB key facts statements), which is the middle of the next month here (the end of turn 2). If it is not paid in full, real cards charge interest from each transaction date; the game shows this as one month at 3% on the whole bill (an AED 2,000 bill with AED 200 left adds about AED 60). The minimum is 5% of the bill, at least AED 100; under it, a late payment fee (Emirates NBD and Standard Chartered UAE: AED

241.50, AED 240 here; Emirates NBD credit card Key Facts Statement, checked 11 Nov 2026). This is a conventional card; Islamic cards charge a fixed monthly fee instead of interest.

- **Cash advance:** a fee of the higher of AED 105 and 3% (many UAE banks charge about 3%, minimum AED 100 or more) and one month of interest at once, since it has no interest-free period.
- **Interest-free period:** applies only if the previous statement was paid in full (Emirates NBD and FAB key facts; the Emirates NBD KFS above: no interest-free period when the previous month's balance was not cleared in full). In the game, the bill after a missed one pays 3% interest even if it is paid in full; paying it in full brings the free period back.
- **Payroll delay (month 8):** the salary comes two weeks late (an example). Late salaries happen; the Wage Protection System tracks every payroll, and from 1 June 2026 Ministerial Resolution 340 is reported to set the 1st of the month as the deadline and end the earlier grace days (DLA Piper, Clyde & Co; the resolution text itself not yet read). Rent needs cash or a transfer, so what the emergency fund and then savings cannot pay of the rent is a cash advance; food and transport go on the card.
- **First month:** the first salary is pro-rated (24 of 30 days); the room deposit (AED 1,800, about one month of rent) is refundable and counts in net worth once paid.
- **Card limit:** AED 10,000, lower than real first-job limits (often 2 to 3 salaries) so one short year can show the spiral; the end screen says so.
- **Bills left unpaid** stay owed with an AED 100 late fee and come back next month (example value; utilities and telecoms charge late fees and can cut the service). They do not move to the card.
- **Pay-later:** 4 parts of AED 900, part 1 on purchase, then one part from each of the next three salaries on payday. If the cash is short, the part is charged to the card on file, with no late fee: Tabby charges AED 15, 30 and 60 only when a payment fails (Tabby help: missed payments, checked 11 Nov 2026).
- **Agent fee:** AED 1,100, about 5% of a year's rent (Bayut) on a shared room of AED 21,600 a year.
- **Living costs** of AED 4,300 for a single young worker: shared room 1,800, food 1,000, transport 800, phone 200, utilities and internet 300, other 200. The phone, power and DEWA bills in the year are the part above that base (roaming and data, summer air conditioning). The room deposit, traffic fine, car registration, Salik, utility bills, the flight fare and the extra baggage fee are set prices: Compare prices and Negotiate do not work on them (checked by `tools/regress_cards.js` K1b). Surprise cost prices (car AC compressor AED 3,200, emergency root canal AED 2,800) and bills (AED 200 to 450) are rough Dubai levels and need a local check before release.
- **Card spending and Impulse buys:** one Impulse buy (AED 200 if still in hand at turn end) per swipe from month 2 is a game rule built on the studies above, not a measured rate. The worry for a surprise cost on the card (Mood -1 per AED 300 the fund could not cover, at most -8), Mood -3 for breaking into savings, AED 400 of tired spending in a tired month and AED 1,500 at a burnout, and the side gig (AED 450, one a month) are example values. Tools/regress_lesson.js plays 11 styles (including paying only the card minimum, side gigs with no planned joy and leaving small bills late) on 240 class codes and fails if any ends above steady habits on average.
- **Save ahead:** money set aside for the flight home only (AED 400 a card, up to the flight price). After the flight, money left over goes back to savings and the cards leave the deck.
- **Profit-sharing fund:** an example of a Shariah-compliant (mudaraba or musharaka style) fund, not a real product. Example returns: about 0.1 to 1% a month, a loss of 5 to 8% in the

month 8 dip, then about 1.6 to 2.2% a month (values vary by code). Selling early costs 2%. The game says nothing is promised.

- **Zakat:** the game checks savings, the fund and the emergency fund, less debts due now (card debt, pay-later instalments, late bills), at year end against a nisab of AED 42,266 (85 g of 24K gold at AED 497.25 a gram, UAE gold price on 10 Nov 2026; a constant in data.js, to check again before release). A first-year salary stays below the gold nisab, so players meet the "not due" case; the hawl (a lunar year) counts only from the day wealth reaches the nisab, and the game says so. Some scholars use the silver nisab (595 g of silver), which is much lower; the game says so and points players to a scholar or the UAE Zakat Fund calculator. The quick guess on zakat accepts the silver nisab answer as "right by that view". The Shariah review in phase P2 will confirm this wording.
- **Qard hasan:** the friend's loan returns only the same amount after two months; the cash advance option costs the cash advance fee (the higher of AED 105 and 3%) plus AED 45 of interest at once.
- **Months** follow the 2027 calendar (Ramadan from early February, Eid al-Fitr early March, summer in June and July, November deals, the Dubai Shopping Festival in December). Ramadan and Eid are planned joy, never enemies.